

Nonlinear-Map Manifold POD Operator Inference

A nonintrusive closure-map extension of polynomial-manifold OpInf

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Abstract

Polynomial-manifold Operator Inference (OpInf) enriches a low-dimensional linear modal approximation by expressing secondary modal coordinates as polynomial functions of the primary coordinates. This work considers the corresponding nonlinear-map construction

$$\mathbf{u}(t; \boldsymbol{\mu}) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}),$$

where the secondary coordinate map $\mathcal{N}$ is learned from data by a Gaussian process or neural network. The construction is motivated by intrusive closure-map projection methods, but is placed here in the nonintrusive OpInf setting, where the reduced vector-field library must be specified explicitly from data.

For quadratic full-order dynamics, the modal substitution shows that the quadratically complete reduced library in the augmented coordinates $[\mathbf{q}^T, \mathbf{z}^T]^T$, $\mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu})$, contains $\mathbf{q} \otimes \mathbf{q}$, $\mathbf{q} \otimes \mathbf{z}$, and $\mathbf{z} \otimes \mathbf{z}$. We compare this quadratic-complete model with two cheaper nested alternatives: an augmented-linear model in $[\mathbf{q}, \mathbf{z}]$ and a primary-quadratic model that keeps $\mathbf{q} \otimes \mathbf{q}$ but omits the mixed and secondary quadratic blocks. Numerical tests on a Korteweg–de Vries benchmark and a parametric inviscid Burgers problem show that the learned nonlinear map can greatly improve accuracy over linear-subspace OpInf and polynomial MPOD. The Burgers results also show a clear tradeoff: the quadratic-complete library is the most accurate but, in its uncompressed form, is too expensive online; the augmented-linear variant gives the best accuracy-cost compromise in the present experiments.

1 Introduction

Projection-based reduced-order models are most effective when the solution manifold is well approximated by a low-dimensional linear subspace. For transport-dominated problems this assumption can be restrictive: shocks, fronts, and coherent waves often lead to slowly decaying Kolmogorov n -widths. Nonlinear-manifold reduced models address this issue by replacing a purely linear approximation with a low-dimensional nonlinear embedding.

Geelen, Balzano, Wright, and Willcox proposed a nonintrusive OpInf framework in which the nonlinear manifold has polynomial modal structure

$$\mathbf{s}(t) \approx \mathbf{s}_{\text{ref}} + \mathbf{V}\hat{\mathbf{s}}(t) + \bar{\mathbf{V}}\boldsymbol{\Xi}\mathbf{g}_p(\hat{\mathbf{s}}(t)), \quad (1)$$

and the reduced operators are inferred from data after deriving the algebraic structure induced by this manifold [3]. This paper asks a closely related question: can the polynomial secondary-coordinate map in (1) be replaced by a learned nonlinear map while preserving the OpInf structure?

The question is also connected to closure-map projection methods. Quadratic manifold projection [4], neural-network-augmented projection [5], and GPR/RBF latent-space closure models [6]

all exploit the idea that unresolved modal coordinates can be predicted from resolved coordinates. Those methods are intrusive: after the closure map is chosen, the governing equations can be projected directly. In nonintrusive OpInf the full residual is not evaluated, so the induced reduced library must be selected explicitly. The main contribution here is to spell out that library for a learned nonlinear secondary map and to test the resulting models on two benchmarks.

The remainder of the manuscript follows the structure of [3]. Section 2 derives the nonlinear-map manifold and the associated OpInf libraries. Section 3 defines the KdV and Burgers problems. Section 4 presents a compact numerical comparison. Appendix A collects supplementary diagnostics: feature counts, online timings, an extrapolatory visualization, and the MPOD order sweep.

2 Nonlinear-map manifold OpInf

2.1 State approximation

Let $\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the semidiscrete full-order state. We split a POD basis into primary and secondary modes,

$$\mathbf{V} \in \mathbb{R}^{N \times r}, \quad \bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{r}}, \quad \mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0},$$

and approximate the state by

$$\mathbf{u}(t; \boldsymbol{\mu}) \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}}\mathbf{z}(t; \boldsymbol{\mu}), \quad \mathbf{z}(t; \boldsymbol{\mu}) = \mathcal{N}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}). \quad (2)$$

Only $\mathbf{q}(t; \boldsymbol{\mu}) \in \mathbb{R}^r$ is evolved online. The secondary coordinates $\mathbf{z} \in \mathbb{R}^{\bar{r}}$ are recovered algebraically. In this sense (2) is a nonlinear-map analogue of MPOD: MPOD uses $\mathbf{z} = \boldsymbol{\Xi}\mathbf{g}_p(\mathbf{q})$, whereas NM-MPOD uses a learned map.

2.2 Quadratic projected algebra

Consider first a quadratic semidiscrete model without explicit parameter notation,

$$\dot{\mathbf{u}} = \mathbf{c} + \mathbf{A}\mathbf{u} + \mathbf{H}(\mathbf{u} \otimes \mathbf{u}). \quad (3)$$

For a linear POD decoder, $\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}$, substituting into (3) and projecting with $\mathbf{V}^T$ yields

$$\dot{\mathbf{q}} = \hat{\mathbf{c}} + \hat{\mathbf{A}}\mathbf{q} + \hat{\mathbf{H}}(\mathbf{q} \otimes \mathbf{q}). \quad (4)$$

This is the standard quadratic OpInf library.

For the nonlinear-map decoder (2), the chain rule gives

$$\dot{\mathbf{u}} = \mathbf{V}\dot{\mathbf{q}} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}}{\partial \mathbf{q}}(\mathbf{q}, \boldsymbol{\mu}) \dot{\mathbf{q}}.$$

Under the modal projection used in MPOD-OpInf, the tangent contribution vanishes because $\mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}$, so $\mathbf{V}^T \dot{\mathbf{u}} = \dot{\mathbf{q}}$. The right-hand side, however, changes. Expanding the quadratic term gives

$$\begin{aligned} \mathbf{u} \otimes \mathbf{u} &= \mathbf{u}_{\text{ref}} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \mathbf{u}_{\text{ref}} + \mathbf{u}_{\text{ref}} \otimes \bar{\mathbf{V}}\mathbf{z} + \bar{\mathbf{V}}\mathbf{z} \otimes \mathbf{u}_{\text{ref}} \\ &\quad + \mathbf{V}\mathbf{q} \otimes \mathbf{V}\mathbf{q} + \mathbf{V}\mathbf{q} \otimes \bar{\mathbf{V}}\mathbf{z} + \bar{\mathbf{V}}\mathbf{z} \otimes \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathbf{z} \otimes \bar{\mathbf{V}}\mathbf{z}. \end{aligned} \quad (5)$$

Thus the reduced dynamics have the split form

$$\dot{\mathbf{q}} = \mathbf{c}_0 + \mathbf{A}_q \mathbf{q} + \mathbf{A}_z \mathbf{z} + \mathbf{H}_{qq}(\mathbf{q} \otimes \mathbf{q}) + \mathbf{H}_{qz}(\mathbf{q} \otimes \mathbf{z}) + \mathbf{H}_{zz}(\mathbf{z} \otimes \mathbf{z}), \quad \mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}). \quad (6)$$

The mixed and secondary quadratic blocks are therefore not optional from the viewpoint of algebraic completeness. They may be regularized, compressed, or removed as model approximations, but their origin is the quadratic full-order vector field combined with the split decoder.

2.3 Nested NM-MPOD libraries

The three libraries considered in the numerical tests are shown in Table 1. The names describe the reduced vector-field library, not the decoder. All three use the same nonlinear-map state approximation (2).

Table 1: Nested NM-MPOD operator libraries for nonparametric quadratic dynamics. For the Burgers problem, the same libraries are augmented by the parameter features in (12).

Variant	Library $\theta(\mathbf{q}, \mathbf{z})$	Interpretation
AL-NM-MPOD	$[1, \mathbf{q}, \mathbf{z}]^T$	Augmented-linear model in $[\mathbf{q}^T, \mathbf{z}^T]^T$. This is the cheapest learned-map closure.
PQ-NM-MPOD	$[1, \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}]^T$	Primary-quadratic model. It retains the usual $\mathbf{q} \otimes \mathbf{q}$ block, but omits the mixed and secondary quadratic terms.
QC-NM-MPOD	$[1, \mathbf{q}, \mathbf{q}_{\text{quad}}, \mathbf{z}, \mathbf{q} \otimes \mathbf{z}, \mathbf{z}_{\text{quad}}]^T$	Quadratic-complete model in the augmented coordinates. This is the direct library implied by (5).

PQ is included as an intermediate ablation. It is not the algebraically complete model, and in the Burgers tests below it does not provide a favorable accuracy-cost tradeoff relative to AL. Its role is diagnostic: it tests whether retaining the primary quadratic block without the mixed and secondary blocks is sufficient.

2.4 Nonintrusive regressions

Let $\mathbf{q}_j = \mathbf{V}^T(\mathbf{u}_j - \mathbf{u}_{\text{ref}})$ and $\bar{\mathbf{q}}_j = \bar{\mathbf{V}}^T(\mathbf{u}_j - \mathbf{u}_{\text{ref}})$ be the primary and secondary coordinates of the training snapshots. A nonlinear secondary map is fit by solving, for example,

$$\boldsymbol{\psi}^* = \arg \min_{\boldsymbol{\psi}} \sum_{j=1}^M \|\bar{\mathbf{q}}_j - \mathcal{N}_{\boldsymbol{\psi}}(\mathbf{q}_j, \boldsymbol{\mu}_j)\|_2^2 + \alpha \|\boldsymbol{\psi}\|_2^2 \quad (7)$$

for a neural network, or by maximizing the marginal likelihood when $\mathcal{N}$ is represented by a Gaussian process.

Once the decoder is fixed, the reduced dynamics are inferred by continuous-time OpInf. With

$$\boldsymbol{\Theta} = [\boldsymbol{\theta}(\mathbf{q}_1, \boldsymbol{\mu}_1) \quad \cdots \quad \boldsymbol{\theta}(\mathbf{q}_M, \boldsymbol{\mu}_M)], \quad \dot{\mathbf{Q}} = [\dot{\mathbf{q}}_1 \quad \cdots \quad \dot{\mathbf{q}}_M],$$

the regularized regression is

$$\mathbf{W}^* = \arg \min_{\mathbf{W}} \left\| \dot{\mathbf{Q}} - \mathbf{W} \boldsymbol{\Theta} \right\|_F^2 + \|\mathbf{W} \boldsymbol{\Lambda}\|_F^2. \quad (8)$$

The online ROM is then

$$\dot{\mathbf{q}} = \mathbf{W}^* \boldsymbol{\theta}(\mathbf{q}, \boldsymbol{\mu}), \quad \mathbf{u}_{\text{ROM}} = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}). \quad (9)$$

3 Benchmark problems

3.1 Korteweg–de Vries equation

The first test is the periodic Korteweg–de Vries equation

$$s_t + \alpha s s_x + \beta s_{xxx} = 0, \quad x \in [-\pi, \pi), \quad t \in [0, 1], \quad (10)$$

with $\alpha = 4$, $\beta = 1$, periodic boundary conditions, and

$$s(x, 0) = 1 + 24 \operatorname{sech}^2(\sqrt{8}x).$$

The full model uses a Fourier pseudospectral discretization with $N = 256$ grid points and exponential time differencing Runge–Kutta time integration with $\Delta t = 2 \times 10^{-4}$. The interval $t \in [0, 0.2]$ is used for training and $t \in (0.2, 1]$ for prediction.

3.2 Parametric inviscid Burgers equation

The second test is the parametric inviscid Burgers problem on $x \in [0, 100]$,

$$u_t + \left(\frac{u^2}{2} \right)_x = 0.02 \exp(\mu_2 x), \quad u(x, 0) = 1, \quad (11)$$

with left inflow flux $f(u(0, t)) = \mu_1^2/2$ and

$$\boldsymbol{\mu} = (\mu_1, \mu_2)^T, \quad \mu_1 \in [4.25, 5.50], \quad \mu_2 \in [0.015, 0.030].$$

The high-dimensional model uses $N = 5000$ finite-volume cells and backward Euler time integration with $\Delta t = 0.05$ for 500 steps. The nonintrusive parametric library uses

$$\boldsymbol{\eta}(\boldsymbol{\mu}) = [\mu_1 \quad \mu_2 \quad \mu_1^2 \quad \mu_1 \mu_2 \quad \mu_2^2]^T. \quad (12)$$

Thus, for example, the Burgers QC library is

$$\boldsymbol{\theta}_{\text{QC}}(\mathbf{q}, \boldsymbol{\mu}) = \begin{bmatrix} 1 \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \\ \mathbf{q} \\ \boldsymbol{\eta}(\boldsymbol{\mu}) \otimes \mathbf{q} \\ \mathbf{q}_{\text{quad}} \\ \mathbf{z} \\ \mathbf{q} \otimes \mathbf{z} \\ \mathbf{z}_{\text{quad}} \end{bmatrix}, \quad \mathbf{z} = \mathcal{N}(\mathbf{q}, \boldsymbol{\mu}).$$

Training uses the 3×3 parameter grid

$$\mu_1 \in \{4.25, 4.875, 5.50\}, \quad \mu_2 \in \{0.015, 0.0225, 0.030\}.$$

The interpolatory test points are $(4.56, 0.019)$, $(4.75, 0.020)$, and $(5.19, 0.026)$. We also report one extrapolatory point, $\boldsymbol{\mu}^{(3)} = (4.00, 0.033)$, located outside the training box. No test trajectory is used to construct the POD basis, fit the nonlinear map, or infer the reduced operators.

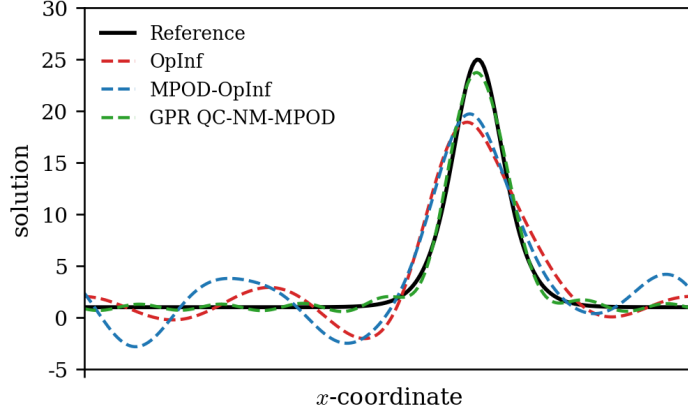
4 Numerical results

4.1 KdV

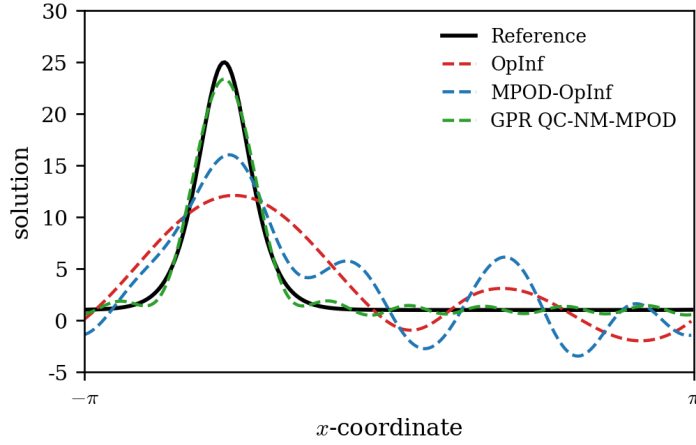
The KdV experiment uses $r = 5$ primary coordinates and $\bar{r} = 9$ secondary coordinates for the learned-map models. Table 2 compares standard OpInf, MPOD, and the three GPR-NM-MPOD libraries. The main effect is the learned secondary map: all three GPR-NM-MPOD variants reduce the prediction error by about a factor of five relative to MPOD. The extra PQ and QC quadratic blocks do not materially improve this KdV case once regularized.

Table 2: KdV relative state errors. Training uses $t \in [0, 0.2]$; prediction uses $t \in (0.2, 1]$.

Model	Full window	Prediction window
Standard OpInf, $r = 5$	5.33×10^{-1}	5.37×10^{-1}
MPOD-OpInf, $r = 5, p = 2$	4.44×10^{-1}	4.48×10^{-1}
GPR AL-NM-MPOD, $r = 5, \bar{r} = 9$	8.73×10^{-2}	8.77×10^{-2}
GPR PQ-NM-MPOD, $r = 5, \bar{r} = 9$	8.73×10^{-2}	8.77×10^{-2}
GPR QC-NM-MPOD, $r = 5, \bar{r} = 9$	8.73×10^{-2}	8.77×10^{-2}



(a) $t = 0.2$



(b) $t = 1$

Figure 1: KdV solution snapshots at the end of the training window, $t = 0.2$, and at the final prediction time, $t = 1$. All reduced models use primary dimension $r = 5$; the learned-map model uses $\bar{r} = 9$ secondary coordinates.

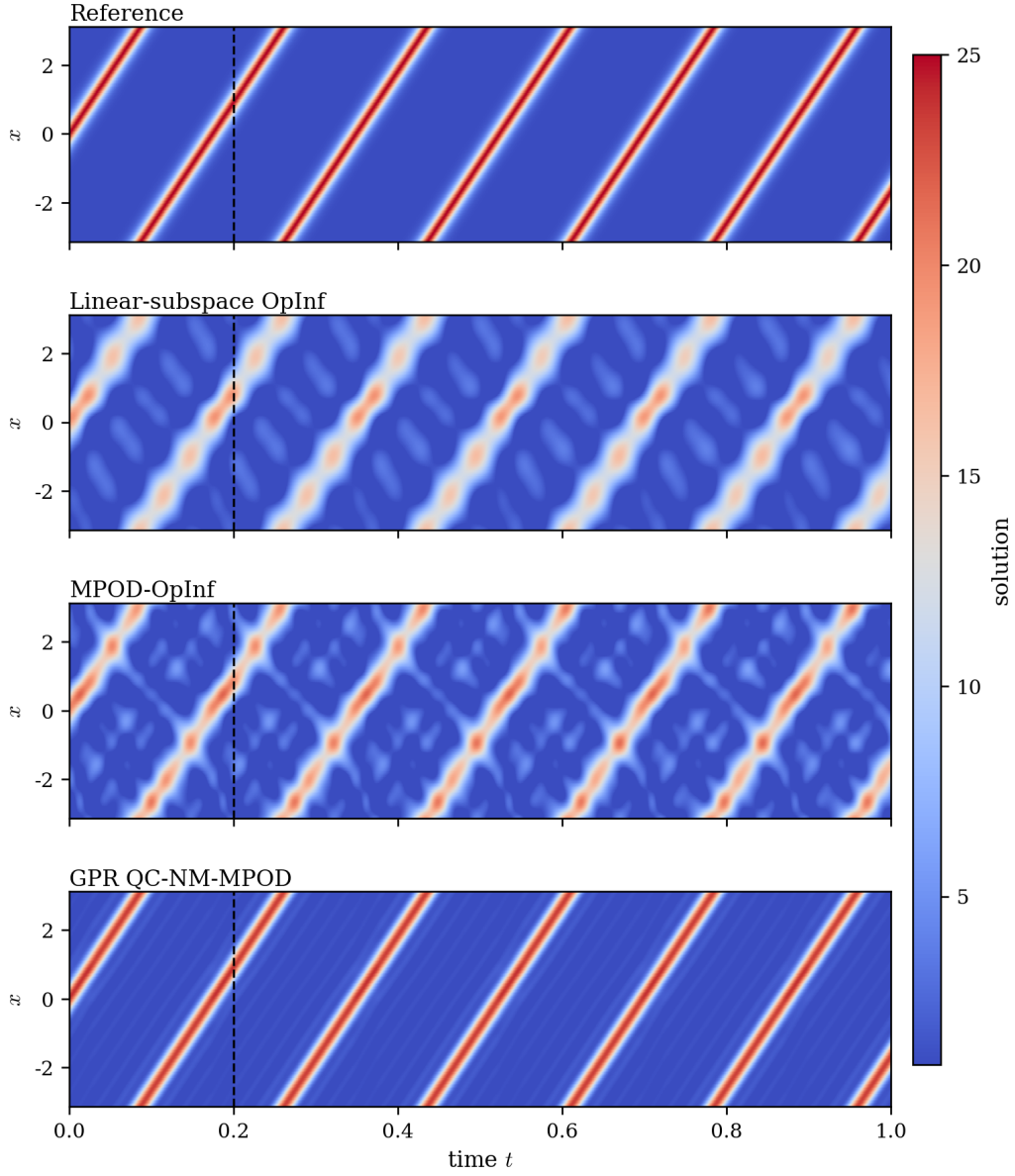


Figure 2: KdV space-time comparison on a common color scale. The dashed line marks the end of the training interval.

For this KdV test, the learned secondary-coordinate map is the decisive ingredient: GPR-NM-MPOD preserves the transported coherent structures through the prediction window, while standard OpInf and polynomial MPOD remain limited by the low-dimensional linear coordinates.

4.2 Parametric Burgers

For Burgers, the retained models evolve $r = 10$ primary coordinates. The ANN-NM-MPOD decoder reconstructs $\bar{r} = 133$ secondary coordinates, for $r + \bar{r} = 143$ POD coordinates in the nonlinear manifold. Hyperparameters are selected by a common training-trajectory validation protocol before evaluating the test points. The comparison is therefore made at the same online state dimension: standard OpInf, MPOD, and NM-MPOD all evolve $\mathbf{q} \in \mathbb{R}^{10}$. The models differ in how they reconstruct unresolved content and in the feature library used to infer $\dot{\mathbf{q}}$, while the training and test trajectories are identical across methods.

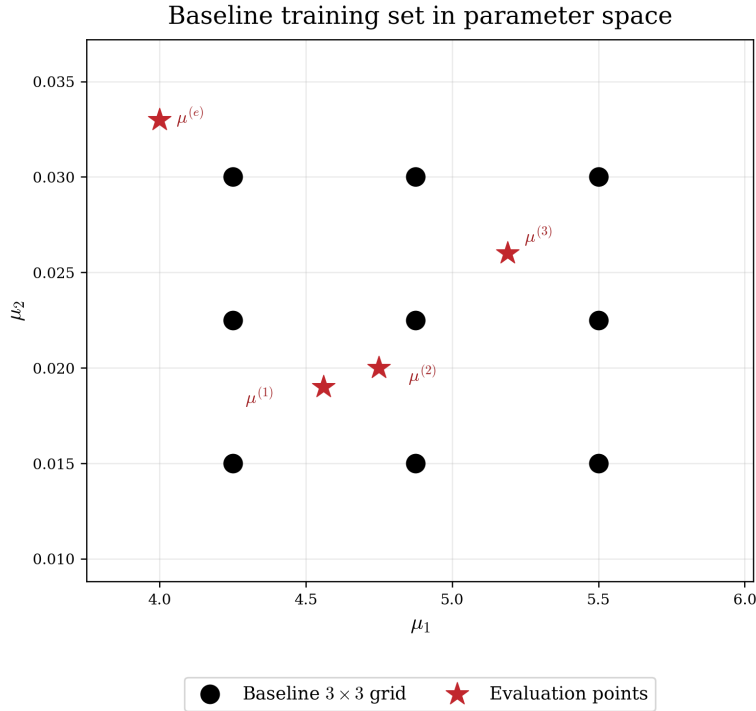


Figure 3: Burgers parameter domain. Black circles are the baseline 3×3 training grid. Red stars are held-out evaluation points, including the extrapolatory stress test.

Table 3 gives the main result in a compact form. The learned-map models are much more accurate than standard OpInf and MPOD at the interpolatory test points. QC is the most accurate, but its uncompressed quadratic library is not computationally viable in this implementation. AL is less accurate than QC but substantially cheaper and is the practical recommendation from this set of tests. PQ is not attractive here: it is slower and less accurate than AL, which suggests that adding the primary quadratic block without the mixed and secondary quadratic terms is not the right compromise for this Burgers configuration.

Table 3: Parametric Burgers relative state errors. “Interp.” is the mean over the three held-out interpolatory parameters. “Extra.” is the error at $\mu^{(3)} = (4.00, 0.033)$.

Model	Interp. error	Extra. error
Standard OpInf	8.99×10^{-2}	7.57×10^{-2}
Induced MPOD, $p = 2$	8.23×10^{-2}	9.20×10^{-2}
ANN AL-NM-MPOD	9.32×10^{-3}	2.47×10^{-2}
ANN PQ-NM-MPOD	1.34×10^{-2}	2.35×10^{-2}
ANN QC-NM-MPOD	5.53×10^{-3}	2.35×10^{-2}

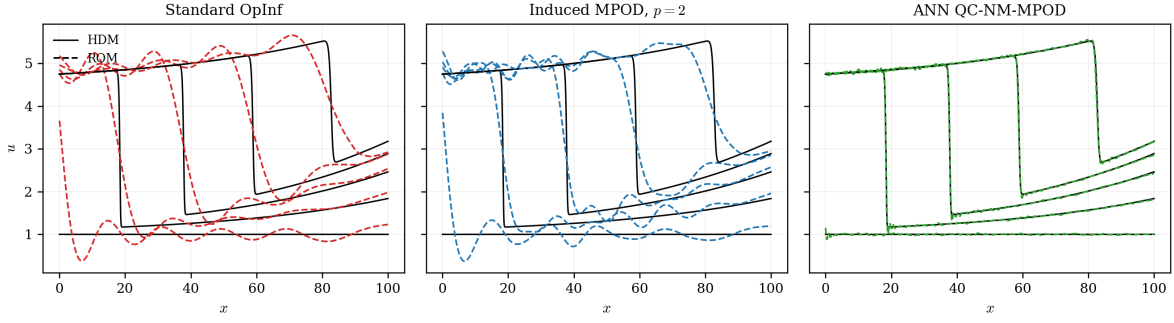


Figure 4: Representative Burgers solution profiles at $\mu = (4.75, 0.020)$. Gray curves are the high-dimensional solution; colored dashed curves are ROM predictions at the same times.

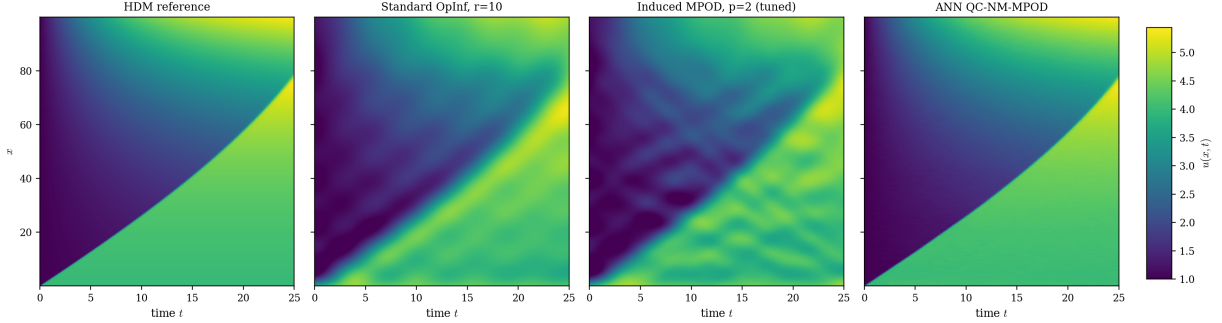


Figure 5: Burgers extrapolatory test at $\mu^{(3)} = (4.00, 0.033)$. The panels show the high-dimensional solution, standard OpInf, induced MPOD with $p = 2$, and ANN QC-NM-MPOD.

For the Burgers test, the same conclusion holds in a parametric setting: at fixed online dimension $r = 10$, the ANN map supplies the missing secondary coordinates and produces substantially better interpolation and extrapolation than the linear-subspace and polynomial-manifold baselines.

5 Discussion

The numerical evidence supports three conclusions. First, the learned secondary map is the dominant source of improvement in both benchmarks. On KdV, AL, PQ, and QC are nearly indistinguishable after regularization; on Burgers, all three ANN-NM-MPOD variants are substantially more accurate than standard OpInf and polynomial MPOD.

Second, algebraic completeness and deployability are different criteria. QC is the only state-quadratically complete library induced by (5). It is therefore the right reference model for asking whether the mixed and secondary quadratic blocks matter. In the present Burgers implementation those blocks improve accuracy but dominate online cost. This makes QC an accuracy reference, not yet a deployable reduced model.

Third, the intermediate PQ model is not a convincing compromise for Burgers. It adds the primary quadratic block to AL but omits the terms that make the augmented quadratic expansion complete. The result is neither the cheapest nor the most accurate model. For the current data, AL is the useful low-cost learned-map closure, while QC identifies the accuracy ceiling and the terms that should be compressed or block-regularized in future work.

6 Conclusions

This study extends polynomial-manifold OpInf by replacing the polynomial secondary-coordinate map with a learned nonlinear closure map. The modal primary/secondary split is retained, so the online state dimension remains r , but the decoder can recover information carried by additional POD modes. For quadratic full-order dynamics, substituting this decoder into the quadratic vector field gives a clear algebraic hierarchy: AL, PQ, and QC.

The experiments suggest that the idea is promising but that the complete quadratic library must be handled carefully. The next steps are therefore not to add more unstructured features, but to make the complete library usable: block-regularized inference, compression of the $\mathbf{q} \otimes \mathbf{z}$ and $\mathbf{z}_{\text{quad}}$ blocks, and validation on held-out parameters rather than only held-out times. These steps would bring the nonintrusive NM-MPOD model closer to the intrusive closure-map methods that motivate it, while preserving the data-driven OpInf setting.

A Supplementary diagnostics

This appendix records numerical details that are useful for interpreting the main results but would interrupt the primary narrative.

Table 4: Burgers library size and online cost. The full-order solve time for a 500-step trajectory at $\boldsymbol{\mu} = (4.75, 0.020)$ is 2.16s. Effective degrees of freedom are the ridge-regression trace measure at the selected regularization.

Model	RHS features	Eff. DOF	Rollout time (s)
Standard OpInf	121	≈ 121	0.115
Induced MPOD, $p = 2$	276	93.1	0.299
Induced MPOD, $p = 4$	856	358.3	0.506
Induced MPOD, $p = 5$	1281	497.2	2.40
ANN AL-NM-MPOD	199	195.7	1.47
ANN PQ-NM-MPOD	254	215.4	1.87
ANN QC-NM-MPOD	10495	4002.5	153.5

The cost table explains why nominal feature counts alone are misleading. The QC model has the best accuracy in Table 3, but forming $\mathbf{q} \otimes \mathbf{z}$ and $\mathbf{z}_{\text{quad}}$ at every Runge–Kutta stage dominates the online time. This is the main reason to treat compression and block-regularization as methodological requirements rather than optional optimizations. This mirrors the cost-accuracy point emphasized

in nonlinear-manifold OpInf: a more accurate nonlinear compression does not by itself imply an online speedup, because algebraic complexity may increase.

Table 5: MPOD polynomial-order sensitivity for Burgers. Each row uses the same validation protocol as the main comparison.

MPOD order	RHS features	RK substeps	Interp. mean error	Speedup
$p = 2$	276	1	8.23×10^{-2}	$7.2\times$
$p = 4$	856	1	8.64×10^{-2}	$4.3\times$
$p = 5$	1281	10	9.09×10^{-1}	$0.90\times$

The $p = 5$ row is a warning about validation coverage. It was selected by the same in-training validation procedure as the other models, but it fails at new parameter values. This does not invalidate MPOD; it shows that enriched libraries should be validated by holding out parameter combinations, not only by holding out time segments from the training parameters.

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